

WP-14 Literature Review — The Normative Strand (Run 04)

The Privacy-is-Value Research Programme

2026-08-17

Lit-review runtime · run 04 manifest (normative strand)

Purpose. First-person direction 2026-08-17: bring the anti-commodification pole (Veliz; the Radin market-inalienability lineage) and the collective-privacy literature into the WP-14 corpus, and stress-test the two joins that direction exposes: the inalienability leg of CTR-LR-02, and the carrying of the collective externality into a subject-side valuation. Unlike runs 02/03 this run also returns PLACE-MENT: per-paper citation recommendations for the strand.

Seat configuration

Seat	Count	Role
sweep:*	4	anti-commodification theory · Veliz corpus + reception · collective/group privacy + externalities ·
refute:*	2	inalienable-pricing precedents targeted threat checks (D2/D3): priced+inalienable join; collective-externality join
judge	1	positioning map + threat verdicts + bibliography + citation placements

D4b honest limit (standing). Same model weights across seats; separation is context isolation. D2 phrasing enforced in absence reports.

Run statistics

- Sweep items surfaced: 58 across 4 modalities
- Refuter covering candidates: 16
- Bibliography additions proposed: 31
- Citation placements proposed: 10

Judge headline

Both novelty legs NARROWED: DP-market pricing (Li-Li-Miklau-Suciu 2013; FL-Market 2021) already owns the priced-formal-quantity-with-non-transferred-records join, and the Bergemann line itself (Bergemann-Wang 2021) anticipates the pooling remedy — WP-14's defensible claims are (a) an adversary-relative, time-drifting reconstruction bound $R(t)$ as the priced quantity within a subject-held stock whose inalienability is theorized as a capital-class property rather than a mechanism feature, and (b) the venue-exclusion + social-data-externality decomposition of the observer-subject gap; related work needs a fourth standing objection (anti-commodification, anchored Radin 1987 + Véliz 2020/2024, which PVM partially vindicates structurally) plus a Viljoen/group-privacy collective sentence.

Outputs (this directory)

- `normative_strand_04.md` - positioning map, threat verdicts, citation placements.
- `bibliography_additions_04.md` - new resolvable citations (A4 verifies before tier-A use).
- `_run_data_04.json` - full structured run output.

The normative strand: positioning map, threat verdicts, placements (run 04)

Positioning map: the poles of the personal-data-value debate

anti-commodification

Anchors. Radin 1987 (Market-Inalienability); Roessler 2015 (closest data-specific statement); Véliz 2020 (Privacy Is Power) and Véliz 2024 (The Ethics of Privacy and Surveillance, OUP); Satz 2010 (noxious markets); Custers & Malgieri 2022 and EDPS Opinion 4/2017 (legal-doctrinal twin); Abdulrahim & Famoroti 2019 (noxious-markets applied to data)

Claim. Personal data belongs to a non-market valuation sphere: pricing it is corrosive per se, trade in it should be banned outright (Véliz: data as toxic asset, vote-selling analogy), and EU fundamental-rights doctrine makes the underlying right non-waivable regardless of consent or payment.

PVM relation. PARTIALLY VINDICATES — with Véliz placed precisely. PVM structurally vindicates two of her claims: (1) inalienability of the stock — under R(t)-scoped disclosure the observer can never reconstruct the underlying record and the quality stake is non-transferable by construction, so the sale Véliz wants legally banned is impossible in PVM's construction, a stronger result than a prohibition; this also explains rather than contradicts Custers-Malgieri/EDPS non-waivability. (2) Collective stakes — her claim that one person's data betrays others is carried formally as the social-data externality (Bergemann-Bonatti-Gan 2022) inside the rent decomposition. PVM REJECTS her third claim, that nothing should therefore be priced: scoped disclosure within the retained stock is legitimately priced; Satz's noxious-market criteria fail to bite precisely because the stock never transfers; and formally, prohibition is dominated by bargaining-position remedies (Gu 2024; Acemoglu et al. 2022's market-shutdown result holds only under conditions, partially vindicating the ban pole without generalizing it). Radin is the frame-owner: her shared-not-sold, markets-at-the-periphery middle position is exactly the position PVM occupies — WP-14 can claim only its technical realization, not the structure itself.

propertisation (and its critics)

Anchors. Already engaged: Laudon 1996, Samuelson 2000, Schwartz 2004, Purtova 2015, Prins 2006. New: Cofone 2022 (Beyond Data Ownership), Ritter & Mayer 2018 (licensing-not-ownership), Calabresi & Melamed 1972 (entitlement-rule taxonomy)

Claim. Assign transferable property rights in personal data (pro side); critics: ownership magnifies consent pathologies, cannot reach inferred data, is self-defeating, and data lacks a coherent legal res.

PVM relation. AGREES with the critics and goes further: transferable title is the wrong object because value sits in reconstructable/inferred content, which R(t) prices directly and property title cannot reach (answering Cofone's strongest point). Against Ritter-Mayer, PVM's retention is structural (reconstruction-bounded), not contractual licensing. Against Calabresi-Melamed, PVM's construction occupies a cell the taxonomy lacks natively: an inalienability rule on the stock combined with a price mechanism on scoped flow, where the inalienability is technological rather than a chosen legal rule.

data-as-labour

Anchors. Arrieta-Ibarra et al. 2018; Posner & Weyl 2018

Claim. Data is undercompensated labour; the observer-subject value gap is unpaid wages; remedy is a data wage negotiated by data unions.

PVM relation. DISAGREES on the diagnosis: the gap is market-position rent (venue exclusion + social-data externality per Bergemann-Bonatti-Gan 2022), not unpaid marginal product, so a data wage misprices the problem. PARTIALLY AGREES on the institutional form: unions-as-pooling survives, but as a bargaining-position instrument over scoped disclosure, not a wage claim over an alienated product.

fiduciary intermediation

Anchors. Delacroix & Lawrence 2019 (already engaged); Marinotti 2022 (data trusts lack a coherent property res)

Claim. Data subjects should act through intermediaries bound by duties of loyalty (trusts, cooperatives) rather than transacting bilaterally.

PVM relation. AGREES as the remedy instrument — PVM's remedy IS pooled bargaining via cooperative/fiduciary structures. Marinotti's res critique is absorbed rather than resisted: because the stock is structurally inalienable, PVM's cooperatives pool disclosure rights and bargaining position (which exist) rather than corpus title (which may not), answering the strongest objection to trust constructions.

formal externality economics

Anchors. Bergemann-Bonatti-Gan 2022 (already engaged); Acemoglu et al. 2022 (Too Much Data); Choi-Jeon-Kim 2019; Ichihashi 2021; Jones & Tonetti 2020; Fallah et al. 2024; Gu 2024 and Sun 2026 (preprints); Bergemann & Wang 2021 (tutorial)

Claim. Correlated-inference externalities depress equilibrium data prices and produce excessive collection even under informed consent; the traded object is designed/scoped disclosure; nonrivalry supports selling use of a retained stock; remedies are mediation, bargaining-power reduction, or (under conditions) market shutdown.

PVM relation. AGREES AND BUILDS ON: this line supplies the externality half of PVM's rent decomposition and the formal case that the gap is structural rather than super-additive aggregation. Two attribution duties: Jones-Tonetti already gives the priced-use-of-retained-stock economic form (retention assumed, not enforced — PVM enforces it via $R(t)$); and Bergemann-Wang 2021 shows the pooling/cooperative remedy is anticipated inside the Bergemann line itself, so WP-14's contribution is the subject-side valuation carrying that remedy, not the remedy.

normative collective privacy

Anchors. Viljoen 2021 (relational data governance — highest-priority gap); Taylor, Floridi & van der Sloot 2017 (group privacy); Fairfield & Engel 2015 (privacy as public good); Barocas & Nissenbaum 2014 (inference end-run); Ben-Shahar 2019 (data pollution); MacCarthy 2011 (earliest externality policy statement)

Claim. Privacy interests are supra-individual: data relations create population-level interests that individualist frameworks (propertarian or dignitarian) cannot represent; individual choice/consent fails; remedies are democratic governance, group empowerment, or Pigouvian regulation without subject compensation.

PVM relation. PARTIALLY VINDICATES: PVM concedes the supra-individual interest is real and identifies the social-data externality as its economic shadow, with pooling answering the group rather than only the individual — this meets Viljoen and Fairfield-Engel halfway. DISAGREES with Viljoen's implication that subject-indexed valuation is per se inadequate ($R(t)$ is subject-indexed while the remedy is collective), with Ben-Shahar's no-subject-compensation Pigouvian frame (a foil, since PVM routes the externality's value back to subjects), and answers Barocas-Nissenbaum's end-run directly: $R(t)$ is defined against adversary inference, so leakage through others' data is priced rather than scoped away.

data-as-capital (critical political economy)

Anchors. Sadowski 2019 (When Data Is Capital)

Claim. Data already functions as a capital class — but accumulated by firms through extraction, without consent or compensation.

PVM relation. DISAMBIGUATES: PVM's 'distinct capital class' must be marked off from Sadowski's firm-held extractive capital in one sentence — PVM relocates the stock to the data subject and makes it structurally inalienable, answering the extraction critique rather than repeating the propertisation move Sadowski attacks.

Threat verdicts

Inalienability leg of CTR-LR-02 — NARROWED (high, 85%)

Target. Inalienability leg of CTR-LR-02: PVM as first to price a formal privacy quantity while holding the underlying stock structurally inalienable, pricing scoped disclosure within a retained stock.

Restatement (what survives). PVM is the first to price an ADVERSARY-RELATIVE, TIME-DRIFTING reconstruction bound $R(t)$ as the disclosure price within a SUBJECT-HELD stock, and the first to theorize buyer-side non-reconstructability as structural inalienability of a capital stock — joining the Calabresi-Melamed/Radin inalienability category to a formal pricing metric — rather than treating non-transfer as a mechanism feature (DP-markets), a broker-held architecture (Li et al.), a worst-case adversary-agnostic budget (FL-Market's LDP), or an assumed retention (Jones-Tonetti). The broad wording ('first to price a formal privacy quantity with the stock non-transferred') is NOT defensible and must be dropped: Li-Li-Miklau-Suciu 2013 priced DP-based privacy loss with buyers structurally unable to obtain raw records (arbitrage-freeness IS a reconstruction bound on combined purchases), and FL-Market 2021 adds subject-side structural retention with a priced LDP level.

Basis. Refuter did its coverage duty: two strong candidates verified by fetch (arXiv:1208.5258, arXiv:2106.04384), six search families probed beyond the digest, and a precise absence report — no source prices an adversary-relative or drifting bound, and legal (Custers-Malgieri, Radin) vs formal (DP-markets) literatures each hold exactly one half of the join, a gap the sweeps independently confirmed ('no paper prices disclosure UNDER an explicit inalienability constraint on the stock'). Confidence capped below 90 because the session search budget expired before two planned probes

(retention-framed selling-privacy auctions; identifiability/reconstruction-risk pricing), so the residue rests on a competent but not exhaustive absence.

Externality integration — NARROWED (medium-high, 75%)

Target. Externality integration: carrying the social-data externality / collective privacy (formal: Bergemann-Bonatti-Gan; normative: Véliz, group privacy) into a subject-side capital valuation with a venue-exclusion rent decomposition.

Restatement (what survives). PVM’s defensible claim is the TRIPLE JOIN plus the decomposition: engaging the normative collective-privacy canon (Véliz, Taylor-Floridi, Viljoen) and formal externality pricing inside a subject-side capital valuation, with the observer-subject gap decomposed into venue-exclusion rent plus social-data externality within a structurally inalienable subject-held stock. Every PAIRWISE join is taken and must be conceded: Garrido-Merchán 2025/26 (preprints) join group-privacy normative concern to Pigouvian externality pricing with redistributive compensation; Fallah et al. 2024 (refereed) pay subjects for both local and collectively-imposed privacy losses; Acemoglu et al. 2022 build an externality-adjusted mediated compensation scheme; Fairfield-Engel 2015 join collective privacy to welfare economics in argument-space. Separately, the pooling/cooperative remedy must be re-attributed: Bergemann-Wang 2021 anticipates it inside the formal anchor’s own line, so WP-14 contributes the valuation that carries the remedy, not the remedy.

Basis. No found work performs the triple join or contains a venue-exclusion decomposition, and the nearest hit (Garrido-Merchán) lacks the capital-class framing, $R(t)$, inalienability, and engagement with the normative canon — and is an unrefereed 2026 preprint. Confidence held to 75 because the refuter’s own not-searched list is material: no citation-graph check of works citing both Véliz/Taylor-Floridi AND Bergemann-Bonatti-Gan, no post-2024 SSRN law-and-economics sweep, no non-English literature — exactly where a triple-join candidate would live. The Bergemann-Wang re-attribution duty stands regardless of any further search.

Refuter outputs (covering art + absence reports)

The inalienability leg of CTR-LR-02 — refuter recommendation: NARROWED

- [strong] Chao Li, Daniel Yang Li, Gerome Miklau & Dan Suciu, “A Theory of Pricing Private Data”, ICDT 2013 / ACM Transactions on Database Systems 39(4) 2014 / CACM 60(12) 2017 (2013) — Prices noisy aggregate query answers as a function of accuracy while the buyer can never obtain the raw records by construction; owner micro-payments are tied to a formal DP-based privacy-loss quantity, and arbitrage-freeness is exactly a bound on what a buyer can reconstruct by combining purchased disclosures. This is pricing scoped disclosure within a stock the buyer never receives — the priced+non-transferred join, a decade before WP-14. Narrowing residue: the stock is alienated to a trusted broker (not subject-held), the bound is worst-case DP (not adversary-relative or time-drifting), and non-transfer is a mechanism feature, never theorized as inalienability or a capital class. <https://arxiv.org/abs/1208.5258>
- [strong] Shuyuan Zheng, Yang Cao, Masatoshi Yoshikawa, Huizhong Li & Qiang Yan, “FL-Market: Trading Private Models in Federated Learning”, IEEE BigData 2022 (arXiv:2106.04384, 2021) (2021) — Structural (architectural) inalienability plus priced formal privacy quantity in one mechanism: raw data never leaves the data owners (federated learning), owners locally perturb gradients under local differential privacy, and the perturbation/privacy level is what the auction prices. The subject retains the stock by construction; only scoped, formally-bounded

disclosure is sold. Residue for WP-14: LDP epsilon is a worst-case, adversary-agnostic bound, not an adversary-relative time-drifting $R(t)$; no capital-class or rent analysis. <https://arxiv.org/abs/2106.04384>

- [partial] Guocheng Liao et al. (personalized-DP pricing line), “Pricing Private Data with Personalized Differential Privacy and Partial Arbitrage Freeness”, arXiv:2105.01651 (2021) — Extends the Li-Miklau-Suciu construction to per-subject (personalized) privacy levels — each owner prices her own formal privacy loss while the buyer still receives only perturbed answers. Evidence the priced-formal-quantity-with-non-transferred-records design is a developed literature, not a single paper. Same residue: worst-case DP, broker-held stock, no inalienability theorization. <https://arxiv.org/abs/2105.01651>
- [partial] Charles I. Jones & Christopher Tonetti, “Nonrivalry and the Economics of Data”, American Economic Review 110(9): 2819-2858 (2020) — General-equilibrium precedent for pricing USE of a stock the consumer retains: nonrivalry means consumers sell use broadly while keeping the record, and consumer-held rights approach the social optimum. Covers the ‘priced flow from a retained stock’ economic form and the subject-held allocation, but has no formal privacy quantity and no reconstruction bound — retention is assumed, not enforced structurally. <https://www.aeaweb.org/articles?id=10.1257/aer.20191330>
- [partial] Shota Ichihashi, “The Economics of Data Externalities”, Journal of Economic Theory 196: 105316 (2021) — Formal precedent for the purchased object being designed/scoped disclosure rather than the underlying stock (information design over what the firm learns). Covers the scoped-disclosure-as-priced-object leg; does not hold the stock inalienable and has no reconstruction bound. <https://www.sciencedirect.com/science/article/abs/pii/S0022053121001332>
- [partial] Bart Custers & Gianclaudio Malgieri, “Priceless Data: Why the EU Fundamental Right to Data Protection Is at Odds with Trade in Personal Data”, Computer Law & Security Review 45 (2022) — Covers the inalienable/non-waivable leg from the legal side (GDPR/Charter rights cannot be alienated; any data-as-counter-performance stays revocable), i.e., a non-waivable stock coexisting with data transactions — but supplies no pricing metric, so it covers only one half of the join. <https://www.sciencedirect.com/science/article/pii/S0267364922000309>
- [partial] Margaret Jane Radin, “Market-Inalienability”, Harvard Law Review 100(8): 1849-1937 (1987) — Owns the conceptual category the leg’s novelty claim sits inside: things that may be shared but not sold, with markets permitted at the periphery — exactly WP-14’s middle position, stated normatively. No pricing metric, so it covers the frame, not the join; but it means WP-14 cannot claim the shared-not-sold structure itself, only its technical realization. <https://people.brandeis.edu/~teuber/Radin-Market-Inalienability.pdf>

Absence report (D2). not_found_in_scope(adversary-relative OR time-drifting reconstruction bound as the priced quantity; inalienability theorized as a property of a subject-held capital stock rather than as a mechanism feature or legal rule). Searched beyond the digest: (1) DP-market pricing with arbitrage-free privacy-loss compensation (Li-Li-Miklau-Suciu lineage and successors); (2) data-never-leaves marketplaces pricing DP budgets under federated architectures (FL-Market, Dealer/DP-Dealer, federated data trading, Stackelberg privacy pricing); (3) usufruct/licensing constructions joined to pricing in privacy economics; (4) human-capital inalienability (‘rented not sold’) applied to personal data pricing; (5) market designs declaring ownership inalienable while trading usage rights with DP compensation; (6) data-trust/data-cooperative priced access with privacy budgets. No source was found that (a) prices a bound defined relative to a specific adversary’s inference capability that drifts over time, (b) frames buyer-side non-reconstructability

as structural inalienability of the stock, or (c) combines a formal pricing metric with the Calabresi-Melamed/Radin inalienability category. The legal literature (Custers-Malgieri, EDPS, Radin, Cofone) holds the stock inalienable with no pricing metric; the DP-market literature prices a formal quantity with the records non-transferred but treats that as mechanism design, with the stock typically alienated to a trusted broker (FL-Market is the exception on broker-alienation but uses worst-case LDP). CAVEAT: the session-wide WebSearch cap (200) was reached after 4 of my searches beyond the digest ran; two planned probes (selling-privacy auctions framed around retention; identifiability/reconstruction-risk pricing) could not execute as searches and were partially substituted by WebFetch verification of the top candidates (arXiv:1208.5258, arXiv:2106.04384, arXiv:2111.15415, Li14Theory_TODS PDF).

The externality integration — refuter recommendation: NARROWED

- [strong] Eduardo C. Garrido-Merchan, “Intersectional Data and the Social Cost of Digital Extraction: A Pigouvian Surcharge”, arXiv:2601.08574 [cs.CY], preprint (2026) (2026) — The closest single instantiation of the threat test as stated: a Pigouvian surcharge on data extraction priced formally (monetised mutual-information / entropy reduction over joint intersectional identity distributions), explicitly framed as internalising a group-level privacy externality, with regulators calibrating the surcharge to ‘embed normative judgments directly into market design’ and the mechanism described as ‘a redistributive institutional shield for vulnerable groups’. This literally joins a group-privacy normative concern with formal externality pricing inside a compensation-adjacent instrument (Pigouvian data tax with restitution possibility to affected subjects or collective funds). It does NOT cover the subject-side capital-class framing, the R(t) reconstruction bound, structural inalienability, or the venue-exclusion rent decomposition, and it is an unrefereed preprint from Jan 2026. <https://arxiv.org/abs/2601.08574>
- [partial] Eduardo C. Garrido-Merchan, “An Information-Theoretic Intersectional Data Valuation Theory”, arXiv:2507.14742, preprint (2025) (2025) — The valuation-theory half of the same programme: formal (mutual-information) data valuation defined at the level of intersectional/group identities, explicitly naming a ‘privacy externality’ (firms profit from profiling while users bear hidden social risk and discrimination costs) and positioning value with data subjects / vulnerable groups via a Pigouvian-style surcharge. Together with 2601.08574 this pair is prior art for ‘externality-adjusted data valuation grounded in a collective/group privacy concern’. Gaps: no engagement with the Veliz/Taylor-Floridi normative canon is visible in the abstract, no capital-stock or rent decomposition, no inalienability construction, preprint status. <https://arxiv.org/abs/2507.14742>
- [partial] Alireza Fallah, Ali Makhdoumi, Azarakhsh Malekian, Asuman Ozdaglar, “Optimal and Differentially Private Data Acquisition: Central and Local Mechanisms”, Operations Research, Vol. 72, No. 3 (2024); also EC’22 (2024) — Refereed prior art for ‘externality-adjusted data prices’ inside a compensation instrument: a data-acquisition mechanism that pays privacy-conscious users for two distinct privacy losses — the local loss from their own disclosure and the central loss imposed via the platform’s released estimate (i.e., the collectively-generated leakage) — so payments are explicitly adjusted for a privacy externality channel. Purely formal: no normative collective-privacy engagement, no capital framing, no rent decomposition. <https://pubsonline.informs.org/doi/10.1287/opre.2022.0014>
- [partial] Daron Acemoglu, Ali Makhdoumi, Azarakhsh Malekian, Asuman Ozdaglar, “Too Much Data: Prices and Inefficiencies in Data Markets”, American Economic Journal: Microeconomics, Vol. 14, No. 4, pp. 218-256 (2022) (2022) — Already in the digest but decisive for the for-

mal leg of the threat: shows correlation-leakage externalities depress equilibrium data prices and proposes a mediated data-sharing scheme — a compensation instrument whose transfers are adjusted for the externality — plus conditions under which shutting the market is welfare-improving (partially vindicating the ban pole formally). Covers ‘externality-adjusted data prices’ but not the normative-collective join, the subject-held capital class, or the venue-exclusion/externality rent decomposition. <https://www.aeaweb.org/articles?id=10.1257/mic.20200200>

- [partial] Joshua A. T. Fairfield and Christoph Engel, “Privacy as a Public Good”, *Duke Law Journal*, Vol. 65, No. 3, pp. 385-457 (2015) (2015) — Covers the normative-collective-to-economics bridge half of the integration: privacy as a public good undersupplied by individually rational disclosure, with group-empowerment remedies. It joins the collective-privacy position to welfare economics but stops short of any pricing/valuation instrument, so it covers the join in argument-space, not in instrument-space. <https://dlj.law.duke.edu/article/privacy-as-a-public-good-fairfield-vol65-iss3/>

Absence report (D2). not_found_in_scope(single work joining the normative collective-privacy canon (Veliz, Taylor/Floridi group privacy, Viljoen) with formal externality pricing (Bergemann-Bonatti-Gan / Acemoglu / Choi-Jeon-Kim line) inside a SUBJECT-SIDE capital valuation, and not_found_in_scope(any prior decomposition of the observer-to-subject value gap into venue-exclusion rent plus social-data externality within a subject-held, structurally inalienable stock)). Searched (6 WebSearch queries beyond the digest, plus 2 targeted arXiv fetches, session search budget then exhausted): (1) Pigouvian tax + data externality + subject compensation; (2) data cooperative collective bargaining formal externality compensation models; (3) Bergemann/Wang collective bargaining over data; (4) Fallah-Makhdoumi-Malekian-Ozdoglar data-acquisition payment mechanisms; (5) privacy-externality compensation mechanism design crossed with group/collective privacy valuation; (6) data-dividend externality valuation critiques and (7) collective-consent/data-trust/data-union formal pricing models. The searches surfaced a 2025-26 preprint pair (Garrido-Merchan) that instantiates ‘Pigouvian data tax with group-redistributive compensation grounded in a group-level privacy externality’ — the nearest hit to the threat test as literally worded — and refereed formal work (Fallah et al. 2024; Acemoglu et al. 2022) covering externality-adjusted payments to subjects. No found work performs the triple join (normative collective privacy + formal externality pricing + subject-side capital valuation), and none contains a venue-exclusion rent decomposition or ties externality pricing to a reconstruction-bound/structural-inalienability construction. Not searched due to budget exhaustion: direct citation-graph checks of who cites both Veliz/Taylor-Floridi AND Bergemann-Bonatti-Gan; non-English literature; SSRN-only law-and-economics working papers post-2024.

Citation placements proposed

- **WP-14** • Related work, propertisation subsection — NEW fourth standing objection paragraph (the main site): State the objection the section currently omits: commodification of personal data is corrosive per se — pricing it degrades the person and the good regardless of transaction design (Radin’s personhood account; Roessler’s data-specific moral-limits argument; Véliz’s live ban position, citing the 2024 OUP monograph as the scholarly form alongside the 2020 book; Satz’s noxious-market criteria as the operational test; Custers-Malgieri and EDPS 4/2017 as its legal-doctrinal expression). Then PVM’s reply: structural inalienability CONCEDES the objection’s core — the stock is never for sale and cannot be transferred even in principle, realizing Radin’s shared-not-sold category technically and explaining EU non-waivability

doctrine rather than conflicting with it — while rejecting the ban conclusion, because what is priced is scoped disclosure within a retained stock, and Satz’s noxious-market criteria (weak agency, harmful outcomes) fail to bite when the record never transfers. *Cites*: Radin 1987; Roessler 2015; Véliz 2020; Véliz 2024; Satz 2010; Custers & Malgieri 2022; EDPS Opinion 4/2017; (optionally Abdulrahim & Famoroti 2019 as the market-design-facing articulation)

- **WP-14** · Same subsection, closing collective-privacy sentence(s): Acknowledge the normative collective-privacy position: data relations create supra-individual, population-level interests that individualist frameworks cannot represent (Viljoen), privacy interests attach to groups and profiles (Taylor-Floridi-van der Sloot), and individually rational disclosure undersupplies privacy as a public good (Fairfield-Engel). PVM’s response: the social-data externality (Bergemann-Bonatti-Gan) is the economic shadow of exactly this claim, and the pooling remedy answers the group, not only the individual — so PVM absorbs the collective critique into the rent decomposition rather than opposing it. *Cites*: Viljoen 2021; Taylor, Floridi & van der Sloot 2017; Fairfield & Engel 2015
- **WP-14** · Contributions / novelty statement of the inalienability claim (and wherever CTR-LR-02’s leg is asserted): MANDATORY narrowing per threat verdict 1: concede that pricing a formal privacy quantity while the buyer never receives the records is an established DP-markets design — Li et al. price DP-based privacy loss with arbitrage-freeness acting as a reconstruction bound on combined purchases; FL-Market adds subject-side structural retention with a priced LDP level; Liao et al. show the design line is developed; Jones-Tonetti give the priced-use-of-retained-stock form in general equilibrium (retention assumed). Restate WP-14’s novelty as: (a) the priced quantity is an adversary-relative, time-drifting reconstruction bound $R(t)$, not a worst-case adversary-agnostic budget; (b) the stock is subject-held, not broker-held; (c) non-reconstructability is theorized as structural inalienability of a capital stock — joining the Calabresi-Melamed/Radin inalienability category to a formal pricing metric, a cell neither the legal nor the DP-markets literature occupies. *Cites*: Li, Li, Miklau & Suci 2013/2014; Zheng et al. (FL-Market) 2021; Liao et al. 2021; Jones & Tonetti 2020; Calabresi & Melamed 1972; Radin 1987
- **WP-14** · Related work, propertisation-critique lineage (updating the Laudon/Samuelson/Schwartz/Purtova line): Add the modern critique: Cofone shows ownership proposals collapse into consent-based transfer rules, magnify consent pathologies, and cannot reach inferred data — PVM’s answer is that $R(t)$ prices reconstructable/inferred content directly, which property title cannot. Ritter-Mayer’s licensing-not-ownership construct is the pro-market cousin of retention, but contractual where PVM’s is structural. *Cites*: Cofone 2022; Ritter & Mayer 2018
- **WP-14** · Wherever scoped disclosure / consent scoping is defended against leakage objections: Engage the end-run objection head-on: inference from others’ data circumvents both anonymity and consent, so naive scoping cannot contain what observers learn. PVM’s reply is that $R(t)$ is defined against the adversary’s inference capability, so leakage-through-others is priced into the bound rather than scoped away — the objection is a design input, not a refutation. *Cites*: Barocas & Nissenbaum 2014
- **WP-14** · Rent-decomposition / externality section: Ground the market-position-rent diagnosis in the formal lineage: informed consent still yields excessive collection via information externalities (Choi-Jeon-Kim); correlation leakage depresses equilibrium prices and yields conditions where shutting the market improves welfare — partially vindicating the ban pole formally (Acemoglu et al.); firms exploit externalities via designed disclosure (Ichihashi); refereed precedent exists for externality-adjusted payments to subjects covering both local and collectively-imposed losses (Fallah et al.). Gu’s result — reducing intermediary bargaining

power protects privacy without the welfare loss of prohibition — is the remedy thesis in formal dress (cite as working paper; verify placement before camera-ready). Distinguish Ben-Shahar’s Pigouvian no-subject-compensation frame as a foil: PVM internalises the same externality but routes value to subjects via pooled bargaining. Optionally add a defensive footnote distinguishing the Garrido-Merchán Pigouvian group-privacy surcharge preprint (nearest recent instantiation; lacks capital class, $R(t)$, inalienability, rent decomposition). *Cites:* Choi, Jeon & Kim 2019; Acemoglu et al. 2022; Ichihashi 2021; Fallah et al. 2024; Gu 2024; Ben-Shahar 2019; (footnote: Garrido-Merchán 2026, arXiv:2601.08574)

- **WP-14** · Remedy section (pooling / cooperative-fiduciary instruments): Honest attribution per threat verdict 2: the pooling/collective-bargaining remedy is anticipated within the Bergemann line itself (internalizing the externality by forming a cooperative), so WP-14’s contribution is the subject-side capital valuation that the remedy operates on, not the remedy form. Marinotti’s *res critique* then does positive work: because the stock is structurally inalienable, PVM’s cooperatives pool disclosure rights and bargaining position (which exist) rather than corpus title (which may not) — answering the standing objection to data-trust constructions that Delacroix-Lawrence leave open. *Cites:* Bergemann & Wang 2021 (FAccT tutorial); Marinotti 2022; Fairfield & Engel 2015
- **WP-14** · Capital-class definition (where ‘distinct capital class held by the data subject’ is introduced): One-sentence disambiguation: the data-as-capital reading exists in critical political economy, but there the capital is firm-held and extractive; PVM relocates the stock to the subject and makes it structurally inalienable, answering the extraction critique rather than repeating the propertisation move it attacks. *Cites:* Sadowski 2019
- **WP-04** · Motivation / introduction (single motivating cite, per budget): Motivate the shelf-life framing: static anonymity and consent guarantees are end-run by inference as background data accumulates — the canonical statement that a privacy guarantee’s effective strength is a function of the evolving external data environment, which is exactly the erosion WP-04 formalizes. *Cites:* Barocas & Nissenbaum 2014
- **WP-07** · Related work on composition of purchased/authorized disclosures (only genuinely warranted addition): Cite arbitrage-freeness in DP query pricing as the mechanism-design ancestor of WP-07’s problem: the constraint that a buyer combining multiple purchased disclosures cannot reconstruct more than was paid for is a composition bound over sequential disclosures; WP-07 generalizes from worst-case static DP budgets to leakage-budgeted agents composing under background accumulation. No other sweep item is genuinely warranted for WP-07. *Cites:* Li, Li, Miklau & Suciú 2013/2014

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31 citations. A4 (citation-verifier) re-verifies every entry against a primary record before tier-A use; provenance_confidence is the runtime's label, not A4's.

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